

Project 2: Distributed Microservices Deployment on AWS EC2

Continuation from Project 1: 5 Microservices across 3 Ubuntu EC2 Instances with API Gateway & Role-Based JWT Authentication

Course / Project:	Cloud Computing / Distributed Microservices Project 2
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1. Multi-Instance Architecture & IP Mapping

EC2 Instance	Operating System	Hosted Microservices	Public IPv4 Address
EC2 Instance 1 (Gateway/Reg/Login)	Ubuntu 22.04 LTS	API Gateway (:4000) Registration (:5001) Login (:5002) MongoDB (:27017)	Public: 44.214.139.176 (Private: 172.31.9.169)
EC2 Instance 2 (Admin Service)	Ubuntu 22.04 LTS	Admin Microservice (:5003)	Public: 100.57.160.206 (Private: 172.31.3.233)
EC2 Instance 3 (User Service)	Ubuntu 22.04 LTS	User Microservice (:5004)	Public: 44.192.127.185 (Private: 172.31.2.148)

Task 1: Ubuntu OS Configuration on all 3 EC2 Instances

All 3 EC2 instances were configured with Ubuntu OS. Automated execution of 'setup-ec2.sh' ensured:

1. Ubuntu package manager update and upgrade: 'sudo apt-get update && sudo apt-get

upgrade -y'

2. Node.js (v20 LTS) and npm installation via NodeSource repository

3. Verification of Git installation ('git --version')

Task 2.1: EC2 Instance 1 Public IP

AWS EC2 Management Console screenshot showing EC2 Instance 1 (API Gateway, Registration, Login) running with assigned Public IPv4 address.

[SCREENSHOT PLACEHOLDER: EC2 Instance 1 Public IP]
(Paste screenshot here or save image as screenshots/task2_ec2_1_ip.png)

Task 2.2: EC2 Instance 2 Public IP

AWS EC2 Management Console screenshot showing EC2 Instance 2 (Admin Microservice) running with assigned Public IPv4 address.

[SCREENSHOT PLACEHOLDER: EC2 Instance 2 Public IP]
(Paste screenshot here or save image as screenshots/task2_ec2_2_ip.png)

Task 2.3: EC2 Instance 3 Public IP

AWS EC2 Management Console screenshot showing EC2 Instance 3 (User Microservice) running with assigned Public IPv4 address.

[SCREENSHOT PLACEHOLDER: EC2 Instance 3 Public IP]
(Paste screenshot here or save image as screenshots/task2_ec2_3_ip.png)

Task 4: Updated GitHub Repository & Code Deployment

The project source code in GitHub was updated with the downstream target EC2 Public IPs configured in 'APIGateway_Microservice/index.js'.

GitHub Repository URL: [YOUR_UPDATED_GITHUB_REPO_URL]

Commands executed on EC2 instances:

git clone [YOUR_UPDATED_GITHUB_REPO_URL]

```
cd University_Project-main
npm run install:all
```

Task 5.1: Add a New Record (POST /register/userregister)

Postman executing 'POST http://44.214.139.176:4000/register/userregister' registering a new user record. Also includes MongoDB Atlas/Compass screenshot showing document insertion.

Expected Output / Status: HTTP 201 Created - { 'message': 'User registered successfully', 'user': { ... } }

[SCREENSHOT PLACEHOLDER: Add a New Record (POST /register/userregister)]
(Paste screenshot here or save image as screenshots/task5_1_add_record_postman.png)

Task 5.1 (MongoDB): MongoDB Document Verification for New Record

MongoDB Compass / shell screenshot displaying the newly created user record with hashed password and role 'user'.

[SCREENSHOT PLACEHOLDER: MongoDB Document Verification for New Record]
(Paste screenshot here or save image as screenshots/task5_1_mongodb.png)

Task 5.2: Duplicate Email Registration Rejection

Postman executing 'POST http://44.214.139.176:4000/register/userregister' with the same email ID. Verifies uniqueness constraint.

Expected Output / Status: HTTP 400 Bad Request - { 'error': 'User with this email already exists' }

[SCREENSHOT PLACEHOLDER: Duplicate Email Registration Rejection]
(Paste screenshot here or save image as screenshots/task5_2_duplicate_email.png)

Task 6.1: Login as Admin and Generate JWT

Postman executing 'POST http://44.214.139.176:4000/auth/login' with Admin credentials and role 'admin'. Returns signed JWT.

Expected Output / Status: HTTP 200 OK - { 'message': 'Login successful', 'token': 'eyJhbGciOi...', 'user': { 'role': 'admin' } }

*[SCREENSHOT PLACEHOLDER: Login as Admin and Generate JWT]
(Paste screenshot here or save image as screenshots/task6_1_admin_login_jwt.png)*

Task 6.2: Login as User and Generate JWT

Postman executing 'POST http://44.214.139.176:4000/auth/login' with User credentials and role 'user'. Returns signed JWT.

Expected Output / Status: HTTP 200 OK - { 'message': 'Login successful', 'token': 'eyJhbGciOi...', 'user': { 'role': 'user' } }

*[SCREENSHOT PLACEHOLDER: Login as User and Generate JWT]
(Paste screenshot here or save image as screenshots/task6_2_user_login_jwt.png)*

Task 7.1: Admin + Admin JWT call viewalluser API

Postman executing 'GET http://44.214.139.176:4000/admin/viewalluser' routed by API Gateway to EC2 Instance 2 (100.57.160.206:5003) with Admin JWT.

Expected Output / Status: HTTP 200 OK - { 'message': 'All users retrieved successfully', 'users': [...] }

*[SCREENSHOT PLACEHOLDER: Admin + Admin JWT call viewalluser API]
(Paste screenshot here or save image as screenshots/task7_1_admin_viewalluser.png)*

Task 7.2: User + User JWT call updateprofile API

Postman executing 'PUT http://44.214.139.176:4000/user/updateprofile' routed by API Gateway to EC2 Instance 3 (44.192.127.185:5004) with User JWT.

Expected Output / Status: HTTP 200 OK - { 'message': 'User profile updated successfully', 'user': { ... } }

*[SCREENSHOT PLACEHOLDER: User + User JWT call updateprofile API]
(Paste screenshot here or save image as screenshots/task7_2_user_updateprofile.png)*

Task 8.1: Admin calling User updateprofile API (RBAC Verification)

Postman executing 'PUT http://44.214.139.176:4000/user/updateprofile' using Admin JWT. API Gateway enforces RBAC and rejects request.

Expected Output / Status: HTTP 403 Forbidden - { 'error': 'Access Denied: Role 'user' is required to access this endpoint. Your role is 'admin'.' }

*[SCREENSHOT PLACEHOLDER: Admin calling User updateprofile API (RBAC Verification)]
(Paste screenshot here or save image as screenshots/task8_1_admin_call_user_api.png)*

Task 8.2: User calling Admin viewalluser API (RBAC Verification)

Postman executing 'GET http://44.214.139.176:4000/admin/viewalluser' using User JWT. API Gateway enforces RBAC and rejects request.

Expected Output / Status: HTTP 403 Forbidden - { 'error': 'Access Denied: Role 'admin' is required to access this endpoint. Your role is 'user'.' }

*[SCREENSHOT PLACEHOLDER: User calling Admin viewalluser API (RBAC Verification)]
(Paste screenshot here or save image as screenshots/task8_2_user_call_admin_api.png)*